Measurement Error's Effect on One-Sample *t* Tests Observed by IRT 3-Parameter Logistic Model

Nicholas E. Baird, McNair Scholar, Bachelor of Science in Psychology; David J. Weiss, Ph. D. Psychology, Department of Psychology

Introduction

Core Concepts

- *t* tests measure the difference between means (x_{avg} - μ_{null}) / ($\sigma_{observed}$ /√n).
- Error in the measuring tool can bias the sample means (Caldwell, 2004).
- *p* values characterize the probability that a sample came from a particular population. A .05 value reflects a 5% probability of obtaining the observed results given the null hypothesis (Nuzzo, 2014).
- A null hypothesis states that there is no difference between a measured mean and a hypothesized mean, which is often used in *t* tests.
- Reliability refers to the consistency of measurement for a set of data measured by a particular instrument.

  Although reliability can suggest that a sample is being measured consistently, one has no further insight into the accuracy of the measurement or the distribution of that accuracy within that measurement.

Significance of Study

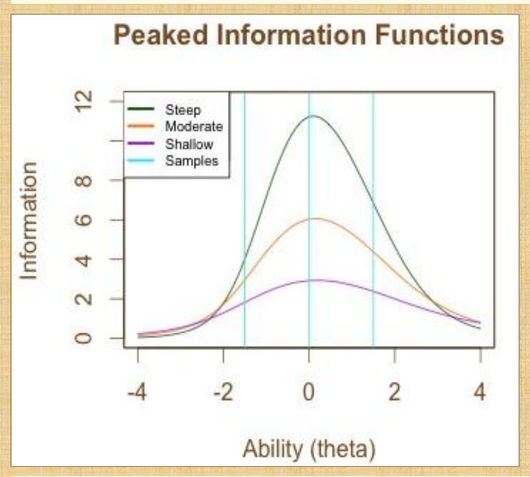
- Measurement error is not thoroughly modeled in current literature.
- Our study is the first exploratory analysis using *Item Response Theory*.

Research Statement

Measurement error will be deconstructed using *Item Response Theory*. The resulting information obtained by the following procedure will analyze a one-sample *t* test and report the patterns seen in Type I error under 18 different conditions of tests and sample groups.

Methodology

1. Monte Carlo simulation of response vectors from 3 distributions of true *θ*, each centered around means of -1.5, 0, and 1.5 (standard deviations = 1).
2. 6 tests were constructed with information functions: steep, moderate, shallow, flat-high, flat-mid, flat-low (Figures 1 and 2).
3. Each test then used an estimation method to measure the 3 true θ distributions (1000 samples of 50 response vectors), producing an estimated simulee score.
4. Maximum Likelihood Estimation (MLE), Bayes Modal Estimation (BME), Weighted likelihood Estimation (WLE) were used as estimation methods.
5. 5. *t* tests, where N = 50, were simulated for each of 1000 samples. The Type I error rate of the estimates (observed) was then compared to that of true *θ.*



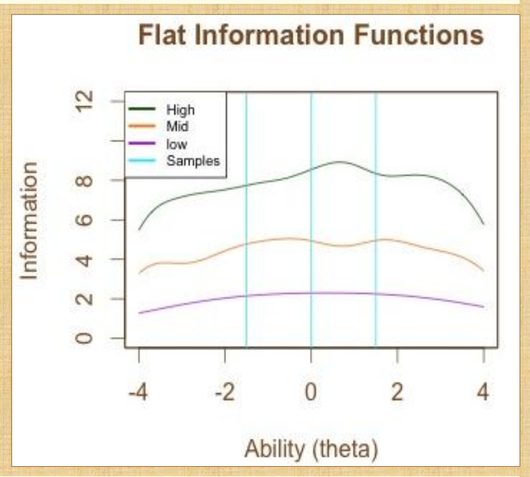
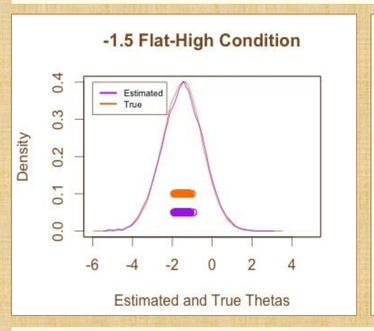


Figure 1 & 2: *Test information functions. The vertical lines identify the locations of the 3 sample true θ distributions used in the study. Left(1): Peaked information tests. Right(2) Flat information tests.*

Results

- In this context, we define measurement error as: *the difference between the error-free true θ data and the estimated θ data.* Every true *θ* distribution had a Type I error rate of .05. This is the expected error rate for a normal distribution.
- Figures 3 and 5 compare estimated *θ*s in relation to true *θ*s. Figures 4 and 6 show Type I error rate for the given conditions.
- Figure 7 compares all Type I errors produced throughout all 18 conditions.
- Figure 8 compares the mean standard error of measurement (SEM) and standard deviation (SD) of the SEM (18 conditions)
- Figure 9 shows the Type I error rates of the peaked tests for MLE, BME, and WLE and the reliability (Cronbach's alpha) associated with each test type. The Type I errors for *all flat tests were extremely similar across conditions*, the mean for these tests was .055.
- Figures 10 and 11 compare the correlation between Type I error rate and SEM and Type I error rate and reliability (Cronbach).



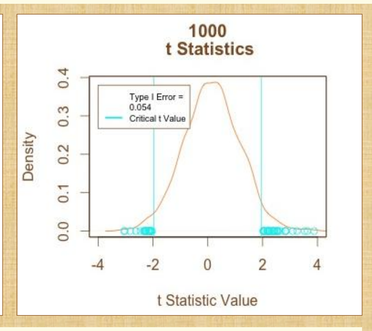
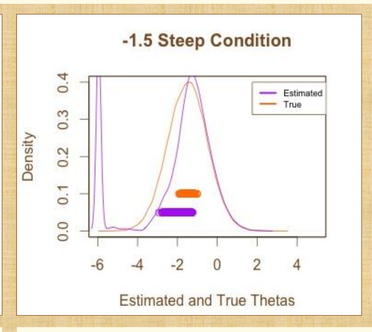


Figure 3 & 4: *Left(3): Distribution of MLE estimated θ and true θ. Right(4): Associated distribution of Type I error.*



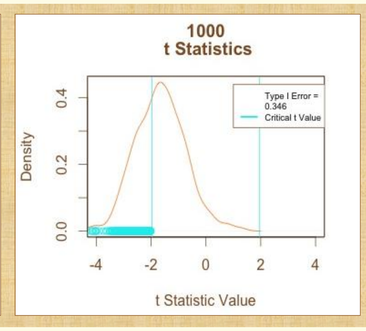
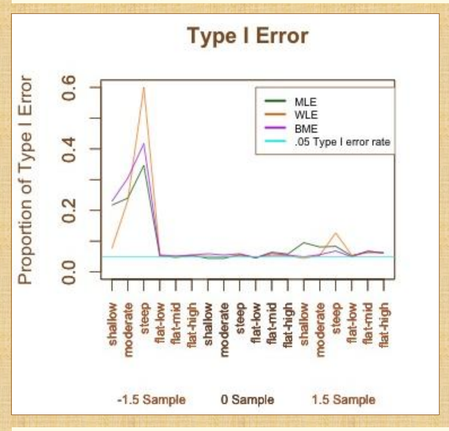


Figure 5 & 6: *Left(5): Distribution of MLE estimated θ and true θ. Right(6): Associated distribution of Type I error.*



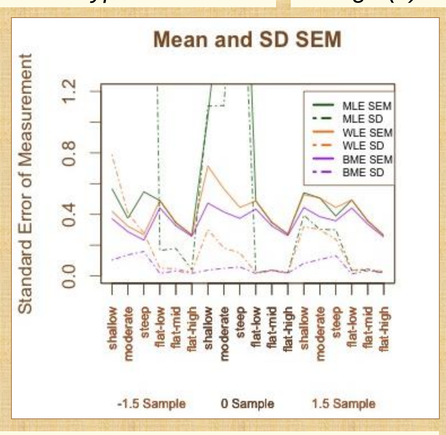
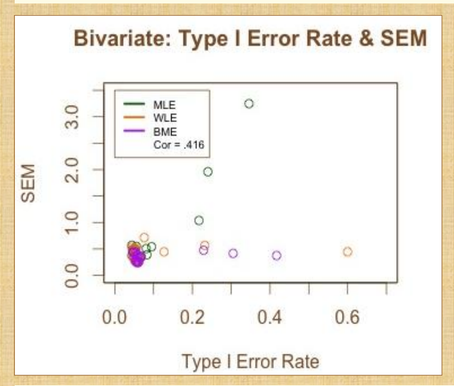


Figure 7 & 8: *Left(7): Type I error rate of all 18 conditions. Right(8): SEM and SD of all 18 conditions.*

Type I Error	Shallow	Moderate	Steep
MLE	0.217	0.240	0.346
BME	0.229	0.305	0.417
WLE	0.076	0.232	0.600
Cronbach's α	0.682	0.768	0.799

Figure 9: *Type I error rates of peaked tests from MLE, BME, and WLE.*



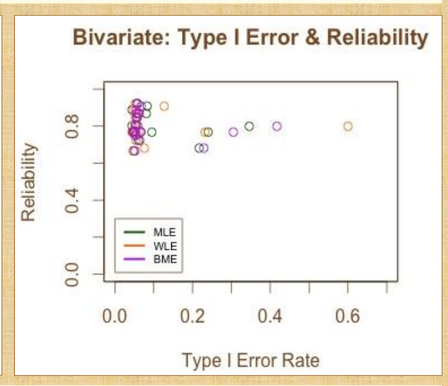


Figure 10 & 11: *Left(10): correlation plot of SEM and Type I error rate of all 18 conditions and of all 4 estimation methods. Right(11): Correlation plot of reliability and Type I error rate for all conditions and methods.*

Discussion

- Peaked conditions can result in an off-target testing situation. This occurs when the distribution of the sample to be measured is not matched by the distribution of the test's measurement parameters.
- Consequently, measurement error had the most substantial impact on Type I error rate when the test information was peaked and approached low information values.
- Reliability has no relationship to Type I error (figure 10).
- Measurement error is a complicated subject matter and our study supports that reliability alone can not determine the magnitude of measurement error or its effect on Type I error.
- A more sophisticated approach to statistical analysis is needed taking into account realistic measurement error.
- *Item Response Theory* has been shown to be adequate in exploring an instrument's impact on Type I error.